

CY-321	Secure Software Design and Development (3 CH)	Fall 2026	CYS
Pre-Requisite: CY-201 Instructor: Dr. Zubair Ahmad Office # G-14, FCSE, GIK Institute, Ext. 2458 Email: zubair.ahmad@giki.edu.pk Office Hours: 9:00 a.m. – 4:30 p.m.			

Course Introduction
The course aims to build essential skills in Secure Software Concepts, Requirements, Design, Implementation/Coding, and Testing. It highlights the security practices that must be integrated across the software development lifecycle. The course offers a thorough overview, addressing the key aspects of people, processes, and technology related to software security defenses.

Course Contents
An overview of techniques and tools for developing secure software with a focus on application security. Key topics include secure software development processes, risk assessment, secure requirements and architectures, static and dynamic code analysis, defensive coding practices, common vulnerabilities, security mechanisms, and secure APIs, secure modern web development, and security testing practices.

Mapping of CLOs and PLOs			
Sr. No.	Course Learning Outcomes (CLOs)⁺	Graduate Attributes (GA)	Bloom's Taxonomy level (Cognitive domain)
CLO 1	Understand software security standards, models, processes and best practices that need to be incorporated throughout the software development lifecycle.	GA-3 (Problem Analysis)	C2 (Understanding)
CLO 2	Analyze and mitigate software vulnerabilities by implementing secure coding practices and automated security testing.	GA-5 (Modern Tool Usage)	C4 (Analyzing)
CLO 3	Apply secure software design principles in developing security solutions for real-world applications.	GA-4 (Design/Development of Solutions)	C3 (Applying)

CLO Assessment Mechanism			
Assessment tools	CLO_1	CLO_2	CLO_3
Quizzes	50%	50%	-
Project	-	-	100%
Midterm Exam	50%	-	-
Final Exam	-	50%	-

Overall Grading Policy	
Assessment Items	Percentage
Quizzes	20%
Project	10%
Midterm Exam	30%
Final Exam	40%

Text and Reference Books
Reference books: - Designing Secure Software: A Guide for Developers by Loren Kohnfelder (2023) - Official (ISC)2 Guide to the CSSLP (latest) - Software Security: Building Security In, 1st Edition by Gary McGraw Additional books: - Ross Anderson. Security Engineering. Wiley. 2nd edition.

- Mathias Paye. Software Security Principles, Policies, and Protection. (January 2019, v0.33)
- Theodor Richardson & Charles Thies. Secure Software Design. Jones & Bartlett Learning. 2013
- Secure by Design - Dan Bergh Johnsson, Daniel Deogun, Daniel Sawano – Manning 2019
 - Architecture Patterns with Python - Harry Percival, Bob Gregory – O'Reilly 2020
 - Crafting Test-Driven Software with Python - Alessandro Molina – Packt 2021

Administrative Instruction

- According to institute policy, 80% attendance is mandatory to appear in the final examination.
- Assignments must be submitted as per instructions mentioned in the assignments.
- In any case, there will be no retake of (scheduled/surprise) quizzes.
- For queries, kindly follow the office hours to avoid any inconvenience.

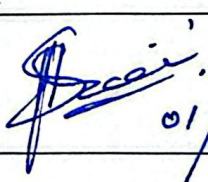
Computer Usage/Software Tool

- Use of data visualization, coding and security testing tools, e.g., Github, Python, JavaScript, Foxhound

Weekly Breakdown

Week 1	• Introduction to Secure Software Concepts • Importance of secure software design • Security in the Software Development Life Cycle (SDLC) • Security concepts integrated into SDLC phases
Week 2	• Risk Management in Software Projects • Identifying, analyzing, and mitigating risks • Introduction to security standards
Week 3	• Security principles and secure design considerations • Designing secure design principles • Research Study
Week 4	• Security Mechanism • Mechanism and Policy • How to regulate access • The K.I.S.S Principle
Week 5	• Common Software Vulnerabilities • Buffer overflow • Exploitation techniques and impacts
Week 6	• SQL injection • Exploitation techniques and impacts
Week 7	• Cross-site Scripting • Exploitation techniques and impacts
Week 8	• Supply Chain Attacks • Exploitation techniques and impacts
Week 9	• Secure Software Testing (1) • Testing concepts: functional, non-functional. • Static analysis
Week 10	• Secure Software Testing (2) • Dynamic analysis • Research Study
Week 11	• API Security • OWASP API Security Top 10

Week 12	• Web security in the real world • Browser Security Mechanism • Building blocks for secure modern web applications.
Week 13	• Taxonomy and classification of adversarial attacks on ML • Threats and exploits
Week 14	• Security and Privacy Compliance in Software Design • Privacy by Design • Data Protection in Software
Week 15	• Emerging trends in secure software • Final Q&A and discussions

Version	Fall 2026
Name of Instructor	Dr. Zubair Ahmad
Instructor's Signature	
Date	24-08-2026
Name of HoD	Prof. Dr. Ghulam Abbas
HoD's Signature	 01/09/26